

Skin Cancer Detection Using Machine Learning and Deep Learning

1. Introduction

Skin cancer is one of the major health concerns associated with abnormal growth of skin cells. Early identification of suspicious skin lesions is important because timely detection can support further clinical examination and treatment. However, identifying whether a skin lesion is benign or malignant from an image can be challenging because lesions may have similar visual characteristics, variations in color and shape, and differences in image quality.

The development of Artificial Intelligence (AI), particularly Machine Learning (ML) and Deep Learning (DL), has provided new approaches for analyzing medical images. Machine Learning algorithms can learn patterns from extracted image features and use those patterns to classify new observations. Deep Learning, particularly Convolutional Neural Networks (CNNs), can automatically learn important visual features directly from images, reducing the need for manually designed features.

This project presents a comparative skin cancer classification system using both traditional Machine Learning and Deep Learning approaches. The objective is to classify skin lesion images into two categories: **Benign** and **Malignant**. For the traditional Machine Learning approaches, Histogram of Oriented Gradients (HOG) is used to extract image features. These features are then used with several classification algorithms, including Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and Support Vector Machine (SVM).

For the Deep Learning approach, a Convolutional Neural Network is developed to learn visual patterns directly from skin lesion images. The CNN uses multiple convolutional and pooling layers followed by fully connected layers for binary classification. The images are resized to 224×224 pixels and normalized before being provided to the network.

The performance of the different models is evaluated using commonly used classification metrics, including **Accuracy, Precision, Recall, and F1-score**. Confusion matrices and comparison graphs are also generated to provide a clearer understanding of the classification performance of the models.

The main purpose of this project is not to replace professional medical diagnosis but to demonstrate how Machine Learning and Deep Learning techniques can be applied to image-based skin lesion classification. The comparison between conventional Machine Learning models and CNN provides an understanding of the advantages and differences between manually extracted image features and automatically learned deep features.

2. Dataset

2.1 Dataset Description

The project uses the **Melanoma Cancer Dataset** downloaded through KaggleHub from the dataset source specified in the notebook:

`bhaveshmittal/melanoma-cancer-dataset`

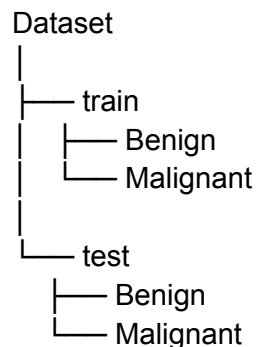
The dataset is downloaded programmatically and the resulting dataset path is used as the base directory for the training and testing data.

The dataset is organized into two main categories:

- **Benign** – non-cancerous skin lesions
- **Malignant** – cancerous skin lesions

The project therefore performs a **binary image classification task**. The class mapping is obtained automatically from the directory structure using Keras' `flow_from_directory()` function.

The dataset is divided into separate training and testing directories:



This organization allows the models to learn from the training images and subsequently evaluate their performance using unseen testing images.

2.2 Image Preprocessing

Different preprocessing strategies are used for the Machine Learning and Deep Learning approaches.

For the CNN model, all images are resized to **224 × 224 pixels**. Pixel values are normalized using a rescaling factor of **1./255**, converting the original pixel range into approximately 0–1.

Data augmentation is also applied to the training images. The implemented augmentation techniques include:

- Rotation
- Width shifting
- Height shifting
- Shearing
- Zooming
- Horizontal flipping

These transformations provide variations of the training images and help the CNN learn more robust visual patterns.

For the traditional Machine Learning models, images are resized to **64 × 64 pixels** and converted to grayscale before HOG feature extraction.

3. Machine Learning Models

3.1 Feature Extraction Using HOG

Traditional Machine Learning algorithms generally require numerical features rather than raw images. Therefore, the project uses **Histogram of Oriented Gradients (HOG)** to extract visual features from the skin lesion images.

First, each image is resized to 64×64 pixels. The image is then converted from RGB/BGR representation into grayscale. HOG analyzes the local intensity gradients and represents the structure and shape information of the image as a numerical feature vector.

The implemented HOG configuration uses:

- Orientations: **9**
- Pixels per cell: **8 × 8**
- Cells per block: **2 × 2**

The extracted HOG features are subsequently standardized using `StandardScaler` before being provided to the Machine Learning classifiers.

3.2 Logistic Regression

Logistic Regression is used as one of the baseline classification algorithms. It learns a relationship between the extracted HOG features and the two target classes.

In this project, Logistic Regression is configured with `max_iter=1000` and trained using the scaled HOG features. The trained model predicts whether a test image belongs to the Benign or Malignant class.

Logistic Regression provides a useful baseline for comparing the performance of more complex classifiers.

3.3 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is another traditional Machine Learning classifier used in the project. KNN classifies an observation based on the classes of its nearest training observations.

The implemented model uses:

```
n_neighbors = 5
```

The classifier is trained using the scaled HOG feature vectors and then used to classify the test images.

3.4 Random Forest

Random Forest is an ensemble Machine Learning algorithm that combines multiple decision trees to produce a final classification result.

The project implements Random Forest using:

```
n_estimators = 200  
random_state = 42
```

The model is trained on the standardized HOG features and generates predictions for the test dataset.

Random Forest is included because ensemble-based learning can capture different patterns in the extracted image features.

3.5 Support Vector Machine

Support Vector Machine (SVM) is also used for classification. The implemented SVM uses an **RBF (Radial Basis Function) kernel**, allowing the classifier to model nonlinear relationships within the extracted feature space.

The implemented configuration is:

```
kernel = "rbf"  
C = 10
```

The SVM is trained using the standardized HOG feature vectors and subsequently used to classify the test images.

4. Deep Learning Model

4.1 Convolutional Neural Network

A Convolutional Neural Network (CNN) is used as the Deep Learning model for skin lesion classification. Unlike the traditional Machine Learning models, the CNN receives the processed images directly and learns important visual representations through its convolutional layers.

The implemented CNN contains three convolutional blocks.

First Convolutional Block

The first block contains:

- 32 convolution filters
- 3×3 convolution kernel

- ReLU activation
- Max pooling

Second Convolutional Block

The second block contains:

- 64 convolution filters
- 3×3 convolution kernel
- ReLU activation
- Max pooling

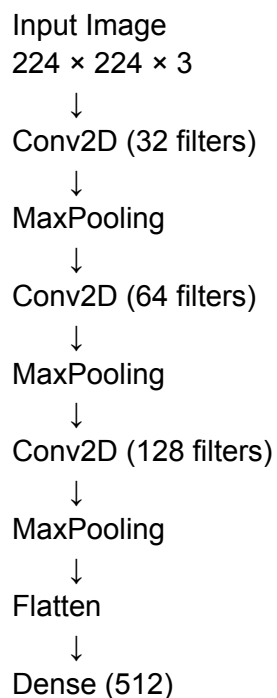
Third Convolutional Block

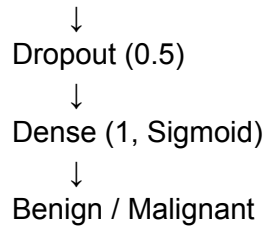
The third block contains:

- 128 convolution filters
- 3×3 convolution kernel
- ReLU activation
- Max pooling

After the convolutional blocks, the feature maps are flattened and passed to a fully connected layer containing **512 neurons**. A dropout layer with a rate of **0.5** is used to reduce overfitting. The final layer contains one neuron with a sigmoid activation function for binary classification.

The architecture can be summarized as:





4.2 CNN Training

The CNN is compiled using the **Adam optimizer** and **binary cross-entropy loss**, which is appropriate for the binary classification problem. Accuracy is used as a training metric.

To improve the training process, two callbacks are used:

Early Stopping

Early stopping monitors validation loss. The implemented model uses a patience value of 2 and restores the best model weights when training stops. This helps prevent unnecessary training after the model stops improving.

Reduce Learning Rate

`ReduceLROnPlateau` reduces the learning rate when the validation loss stops improving. The implementation uses a reduction factor of 0.5, patience of 1, and a minimum learning rate of `1e-6`.

The CNN is trained for a maximum of **5 epochs**, with the training and validation data provided through the image generators.

After training, the model is saved as:

```
skin_cancer_cnn.h5
```

This trained model is subsequently used by the prediction system for classifying new skin lesion images.

5. Model Evaluation

The project evaluates the Machine Learning and CNN models using four primary performance metrics:

Accuracy

Accuracy represents the proportion of correctly classified samples among all tested samples.

Precision

Precision indicates how many samples predicted as positive are actually positive.

Recall

Recall measures how many of the actual positive samples are correctly identified by the model.

F1-Score

F1-score provides a combined measure of precision and recall and is useful when both types of errors are important.

The notebook implements these four metrics using Scikit-learn and generates classification reports for the Machine Learning models.

The CNN is evaluated using the same four metrics. Confusion matrices are also generated for Logistic Regression, KNN, Random Forest, SVM, and CNN to visualize their classification results.

6. Prediction System

The trained CNN is also used as a prediction system for new skin lesion images. The saved CNN model is loaded and a new image is resized to 224×224 pixels and normalized by dividing pixel values by 255. The resulting image is then passed to the CNN for prediction.

The classification rule implemented in the project is:

Prediction > 0.5 → Malignant

Prediction ≤ 0.5 → Benign

The prediction system displays the input image together with its predicted class.

This trained model can also be integrated into a user interface where users upload an image and receive a predicted classification.

7. Summary

The developed system demonstrates the potential application of AI-based image classification to skin lesion analysis. However, the resulting predictions should be considered experimental and should not be treated as a substitute for professional medical diagnosis.